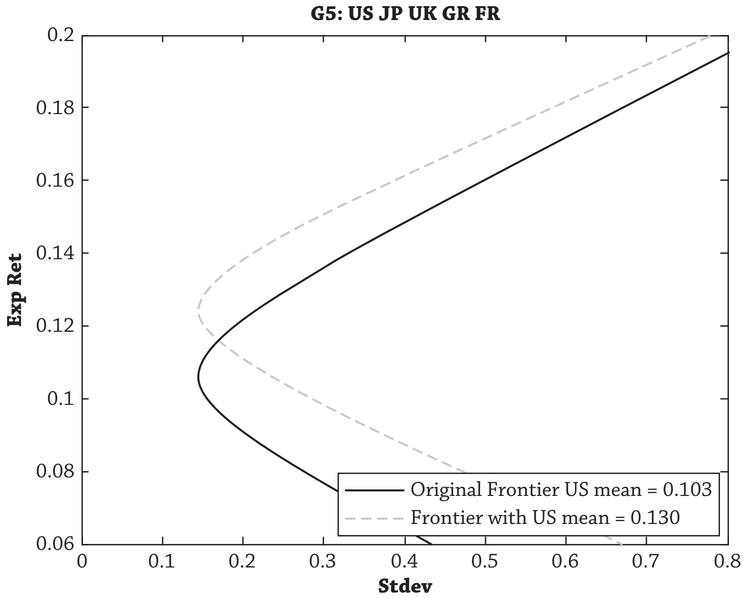

Figure 3.13

gains. (For more general utility functions, see chapter 2.) Unfortunately, we have a dearth of commercial optimizers (none that I know about at the time of writing) that can spit out optimal portfolios for more realistic utility functions, but there are plenty of very fancy mean-variance optimizers. If you insist on (or are forced to use) mean-variance utility. . . .

Use Constraints

Jagannathan and Ma (2003) show that imposing constraints helps a lot. Indeed, raw mean-variance weights are so unstable that practitioners using mean-variance optimization always impose constraints. Constraints help because they bring back unconstrained portfolio weights to economically reasonable positions. Thus, they can be interpreted as a type of *robust estimator* that *shrinks* unconstrained weights back to reasonable values. We can do this more generally if we. . . .

Use Robust Statistics

Investors can significantly improve estimates of inputs by using robust statistical estimators. One class of estimators is *Bayesian shrinkage* methods.[24] These estimators take care of outliers and extreme values that play havoc with traditional classical estimators. They shrink estimates back to a *prior*, or model, which is based on intuition or economics. For example, the raw mean estimated in a

[24] Introduced by James and Stein (1961).